Year 11 Set 4 · Foundation · Lesson 8

Fractions (mock preparation)

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Today I will learn to...

Goals

1. Add and subtract fractions and mixed numbers

2. Multiply and divide fractions and mixed numbers

Do now: retrieval practice

4 mins

1. Gradient of $y = 4x - 2$

   4

2. LCM of 4 and 6

   12

3. Simplify 18/24

   3/4

4. Write $2\frac{1}{3}$ as an improper fraction

   7/3

Objective 1

Add and subtract fractions and mixed numbers

Key idea

Learn

!

Use a common denominator. For mixed numbers, change to improper fractions first.

Objective 1: Add and subtract fractions and mixed numbers

I do: example 1

I do

2/3 + 1/4

1. Common denominator: 12
2. 2/3 = 8/12
3. 1/4 = 3/12
4. 8/12 + 3/12 = 11/12

I do: example 2

I do

5/6 − 1/4

1. Common denominator: 12
2. 5/6 = 10/12
3. 1/4 = 3/12
4. 10/12 − 3/12 = 7/12

I do: example 3

I do

2½ + 1⅔

1. Make improper: 2½ = 5/2, 1⅔ = 5/3
2. Common denominator 6: 15/6 + 10/6
3. = 25/6
4. = 4⅙

Mini whiteboards

We do

1. 3/5 + 1/10

Answer: 7/10

2. 3¼ − 1½

Answer: 1¾

3. 2/3 + 1/4

Answer: 11/12

4. 1⅔ + 2½

Answer: 4⅙

You do: choose your challenge

You do

Bronze

1/3 + 1/6

Silver

7/8 − 2/3

Gold

2¾ + 1⅚

Platinum

A tank is ¼ full. Then 2/5 of the tank's volume is added. What fraction of the tank is empty?

You do: answers

Check

Bronze

1/2

Silver

21/24 − 16/24 = 5/24

Gold

33/12 + 22/12 = 55/12 = 4 7/12

Platinum

5/20 + 8/20 = 13/20 full, 7/20 empty

If you finish early

Extra

Bronze

1/2 + 1/5

7/10

Silver

5/6 − 1/4

7/12

Gold

3½ − 1⅔

1⅚

Platinum

Sam eats 1/3 of a pizza and Jo eats 2/5 of it. What fraction is left?

1 − 11/15 = 4/15

Exam question 1

3 marks

Work out 1¾ + 2⅖. Give your answer as a mixed number.

Space for working

Mark scheme: 35/20 + 48/20 = 83/20 = 4 3/20

Exam question 2

2 marks

3/8 of a class walk to school and 1/3 get the bus. The rest cycle. What fraction cycle?

Space for working

Mark scheme: 1 − 9/24 − 8/24 = 7/24

Objective 1 achieved

Add and subtract fractions and mixed numbers



Objective 2

Multiply and divide fractions and mixed numbers

✓ Achieved: Add and subtract fractions and mixed numbers

Key idea

Learn

!

Multiply: tops × tops, bottoms × bottoms. Divide: keep, change, flip.

Objective 2: Multiply and divide fractions and mixed numbers

I do: example 1

I do

2/3 × 3/5

1. Multiply the numerators: 2 × 3 = 6
2. Multiply the denominators: 3 × 5 = 15
3. = 6/15
4. Simplify (÷ 3): = 2/5

I do: example 2

I do

4/5 ÷ 2/3

1. Keep, change, flip: 4/5 × 3/2
2. Multiply: 12/10
3. Simplify: 6/5
4. = 1⅕

I do: example 3

I do

1½ × 2⅔

1. Make improper: 1½ = 3/2, 2⅔ = 8/3
2. Multiply: 3/2 × 8/3 = 24/6
3. Simplify: 24 ÷ 6
4. = 4

Mini whiteboards

We do

1. $3/4 \times 2/9$

Answer: 1/6

2. $5/6 \div 5/12$

Answer: 2

3. $2/3 \times 9/10$

Answer: 3/5

4. $1\frac{1}{2} \div 3/4$

Answer: 2

You do: choose your challenge

You do

Bronze

$\frac{2}{5} \times \frac{5}{8}$

Silver

$\frac{3}{4} \div \frac{9}{10}$

Gold

$3\frac{1}{3} \times 1\frac{1}{5}$

Platinum

How many $\frac{2}{3}$-litre glasses can be filled from $5\frac{1}{3}$ litres?

You do: answers

Check

Bronze

1/4

Silver

5/6

Gold

$10/3 \times 6/5 = 4$

Platinum

$16/3 \div 2/3 = 8$

If you finish early

Extra

Bronze

$\frac{3}{5} \times \frac{10}{21}$

2/7

Silver

$\frac{4}{9} \div \frac{2}{3}$

2/3

Gold

$2\frac{1}{2} \times 1\frac{3}{5}$

4

Platinum

A ribbon is $4\frac{1}{2}$ m long. How many $\frac{3}{4}$ m pieces can be cut from it?

6

Exam question 1

3 marks

Work out 2¼ ÷ 1½

Space for working

Mark scheme: 9/4 × 2/3 = 18/12 = 1½

Exam question 2

3 marks

Show that $1\frac{2}{3} \times 2\frac{1}{4} = 3\frac{3}{4}$

Space for working

Mark scheme: 5/3 × 9/4 = 45/12 = 15/4 = 3¾

Objective 2 achieved

Multiply and divide fractions and mixed numbers



Real life

Apply

A recipe needs ¾ cup of sugar. You make 2½ times the recipe. How much sugar?

3/4 × 5/2 = 15/8 = 1⅞ cups

Extension: stretch yourself

Stretch

1. A 7½ m ribbon is cut into pieces 3/8 m long. How many pieces?

15/2 ÷ 3/8 = 20 pieces

2. Which fraction is exactly halfway between 1/3 and 1/2?

4/12 and 6/12 → 5/12

Exit ticket: exam questions

Exit

1. Work out 3/4 − 1/3 **(2 marks)**

2. Work out 2½ × 1⅗ **(2 marks)**

3. Work out 1¼ ÷ 5/8 **(2 marks)**

Exit ticket: mark scheme

Mark it

Q1 5/12 [2]

Q2 5/2 × 8/5 = 4 [2]

Q3 5/4 × 8/5 = 2 [2]

Total: _____ / 6

Today I achieved

- ✔ **Add and subtract fractions and mixed numbers**
- ✔ **Multiply and divide fractions and mixed numbers**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Work out 5/6 + 3/8 [2]

2. Work out 4⅕ − 2½ [2]

3. Work out 3/7 × 14/15 [2]

4. Work out 2⅔ ÷ 1⅓ [2]

5. Tia spends ¼ of her money on a ticket and 2/5 of what is left on food. What fraction of her money is left? [3]

Extra practice: Maths Genie → Edexcel question bank → Fractions